

Xinzhe Chen

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EDUCATIONAL BACKGROUND

National University of Singapore (NUS)

Singapore

M.Sc. in Computer Engineering, College of Design and Engineering

Aug 2025 – Jan 2027

- **Overall GPA:** 4.38/5.0

University of Electronic Science and Technology of China (UESTC)

Chengdu, China

B.Eng. in Cyberspace Security, School of Computer Science & Engineering

Sep 2021 – Jul 2025

- **Overall GPA:** 3.73/4.0

Research Interests: Applied cryptography, network security and privacy.

RESEARCH EXPERIENCE

Incremental Verifiable Inference for Large Language Models

-Under submission to ICLR 2027

1st Author (with Asst. Prof. Jiaheng Zhang's team, SoC, NUS)

Mar 2026 – Present

- Designed techniques to improve the efficiency and scalability of incremental verifiable inference for large language models.
- Implemented an end-to-end proof system in Rust for a hundred-millions-parameter scale language model and conducted large-scale experiments on long autoregressive inference.

Distributed zkSNARK via Folding Schemes

-Accepted by Journal of Cryptologic Research

Author (Collaborator: Zesheng Li, IEE, CAS)

Sep 2025 – May 2026

- Designed and implemented a distributed SNARK framework using folding schemes to aggregate distributed sumcheck instances and reduce polynomial-commitment opening overhead.
- Evaluated the Rust implementation against HyperPianist, achieving 4.1–4.9× prover speedup with 8 machines while producing smaller proofs.

Code-based Distributed Polynomial Commitment Scheme

-Under submission to PKC 2027

2nd Author (Collaborator: Zesheng Li, IEE, CAS)

Feb 2026 – Aug 2026

- Designed and implemented a transparent, post-quantum distributed polynomial commitment scheme with linear prover time and polylogarithmic communication.
- Evaluated parallel scalability and prior distributed PCS baselines, achieving at least 13× lower communication while remaining within 1.31× of the fastest prover implementation.

SAS-Based Post-Quantum Signature Public Key Exchange

-Grade A

Course Project (Supervisor: Prof. Marco Tomamichel, CQT, NUS)

Sep 2025 – Apr 2026

- Studied multidimensional cost and security trade-offs of hash-based and other post-quantum digital signatures schemes in practical network environment.
- Designed a Short-Authentication-String (SAS) based key exchange protocol without relying on PKI.

Redactable Blockchain with Secure Trapdoor Key Exchange

- Outstanding Graduation Project

Graduation Project (Supervisor: Assoc. Prof. Hari Krishna Garg, ECE, NUS)

Sep 2024 – May 2025

- Designed a redactable blockchain framework with secure trapdoor key exchange, enabling authorized data modification while preserving blockchain integrity.

INTERNSHIP EXPERIENCE

Huawei 2012 Laboratories, Singapore Research Center

Singapore

Research Intern (Supervisor: Faye Liu)

Sep 2025 – Present

- Research on post-quantum cryptography migration and AI offloading for IEEE 802.11/Wi-Fi standardization.

- Developed platforms for AI-assisted standardization analysis and offensive/defensive simulation.
- Research on formal verification for emerging hybrid-post-quantum protocols from IETF.
- Tracked latest research from WFA, NIST, Bluetooth, and academic conferences.

Beijing Institute of Mathematical Sciences and Applications (BIMSA)

-[Invited Talk](#)

Research Assistant (Organizer: Asst. Prof. Yingjie Zhang)

Dec 2025 – May 2026

- Studied and analyzed the VOLE-in-the-Head framework and its application to constructing the FAEST post-quantum digital signature scheme.
- Investigated how different one-way functions affect signature size and efficiency within the framework.

CAMPUS EXPERIENCE

The Students' Union of SCSE in UESTC

Chengdu, China

Executive President

Sep 2023 – Jul 2024

- Led the student union and was awarded the *Best Student Union President* by UESTC.

HONORS AND AWARDS

Outstanding Graduation Project (*NUSRI*)

May 2025

Outstanding Student Scholarship (*UESTC*)

Dec 2024

Model Student Scholarship (*UESTC*)

Dec 2022, 2023

SKILLS

- **Programming:** Python, Rust, LaTeX
- **Cryptography:** Zero-knowledge proofs, zkSNARKs, PCSs, post-quantum cryptography, blockchain
- **Protocols:** IEEE 802.11 WPA3-SAE/TGbt drafts, Wi-Fi DPP bootstrappings, IETF hybrid PQC PAKE
- **Tools:** Git, Linux, Google Cloud Platform, Overleaf